

Mayatra Vidhi

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SUMMARY

Driven student with a keen interest in physics and health care, seeking a PhD to develop my skills and contribute to research in these areas.

EDUCATION

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| IISER Thiruvananthapuram , BS-MS Integrated (5th Year) | 2022 – 2027 |
| <ul style="list-style-type: none">• Major: Industrial Physics | |
| IIT Madras , BS degree | 2022 – 2025 |
| <ul style="list-style-type: none">• Data Science and Application• Completed foundation coursework (all courses except final Python OPPE); covered statistics, linear algebra, programming, databases, and data science applications. | |

SUMMER INTERNSHIP

CRC 1261 - Kiel University (CAU), Faculty of Engineering, Chair for Integrated Systems and Photonics, Kiel, Germany Collaborative project involving CAU, UKSH, Universitätsklinikum Heidelberg, Fraunhofer ISIT, and IPN. Supervisor: Professor Martina Gerken	May 2026 - August 2026
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- Conducted experimental research on surface-acoustic-wave (SAW) magnetoelastic sensor-based detection of magnetic nanoparticles (MNPs) for biomedical sensing and the early detection of complications following gastrointestinal cancer surgery.
- Performed SAW sensor frequency sweeps and sensor-to-sample distance sweeps using 3D-printed sample holders.
- Investigated MNP detection using biological samples, including turkey-breast tissue and colon cell cultures.
- Performed binding measurements of streptavidin-functionalized MNPs with biotin-coated liposomes using SAW sensors, with complementary measurements using magnetic particle imaging (MPI) and dynamic light scattering (DLS).
- In parallel, prepared MNP dilution series and contributed to fixed-frequency concentration-sweep experiments using Delta-E magnetoelectric sensors.
- Contributed to Delta-E magnetoelectric sensor experiments for MNP detection and prepared a scientific block diagram for an accepted IEEE SENSORS 2026 conference paper.
- Co-authored the accepted IEEE SENSORS 2026 paper "Magnetic Nanoparticle Quantification Using a Delta-E Magnetoelectric Sensor."
- Wrote first-author abstracts for StuFoTech and the CRC 1261 Summer School; delivered a presentation at StuFoTech Co and prepared a poster for the CRC 1261 Summer School.

Outcome: Contributed to SAW and Delta-E-sensor experiments for magnetic nanoparticle detection; gained hands-on experience with MPI, DLS, and biological-sample measurements; co-authored an accepted IEEE SENSORS 2026 conference paper; and presented research at StuFoTech and the CRC 1261 Summer School.

Institute of Biomaterials and Biomolecular Systems, University of Stuttgart
Supervisor: Professor Amirazza Aghakhani

July 2025 - August 2025

- Designed and fabricated PDMS microfluidic devices and hollow silica microrobots
- Operated electromagnetic coil and permanent magnet setups for precise microrobot control.
- Injected and analyzed particle movement in microchannels; evaluated effects of magnetic fields, channel dimensions, and particle size.
- Used MATLAB for experimental control and data analysis
- Technical Skills:
- Microfluidics fabrication, hollow microrobot synthesis (centrifugation, sonication), magnetic manipulation, microscopy, MATLAB analysis.

Outcome: Demonstrated controlled injection and motion of microrobots for biomedical applications; improved understanding of magnetic actuation and fabrication methods, use of FIJI.

School of Veterinary Medicine, University of Surrey
Supervisor: Professor John Joe McFadden

June 2025 - Ongoing

- Conducting remote research on the physical basis of consciousness, focusing on electromagnetic field theories and EEG analysis.
- Identifying and characterizing EEG datasets (normal vs. abnormal) and clarifying criteria used to define normal and pathological activity.
- Using code to analyse EEG data under conventional Maxwell's equations and under extended (complex-valued) Maxwell-type formulations, and comparing how the two frameworks differ in capturing neural field dynamics.
- Regularly reviewing and synthesizing theoretical models and scientific literature on electromagnetic theories of consciousness.
- Technical Skills:
- EEG signal analysis, Python-based computational modelling, theoretical electromagnetism, scientific literature review, remote scientific collaboration.

Outcome: Strengthened understanding of how electromagnetic field formulations applied to EEG data may relate to consciousness, while developing analytical, coding, and remote research skills.

Department of Physics, IIT Roorkee
Supervisor: Professor Akhilesh Kumar Mishra

May 2024 - July 2024

- Optical Bloch Equations (OBEs), modeling atom-light interactions in two-level quantum systems.
- Explored rotating wave approximation, Bloch sphere, and quantum coherence in light-matter systems.
- Used computational methods (Runge-Kutta) for dynamic simulations and analyzed quantum relaxation processes.
- Technical Skills:
- Quantum theory, computational modeling (Python), mathematical analysis, scientific presentation.

Outcome: Produced report on OBEs and their applications in quantum optics and information technology.

SEMESTER PROJECT

Department of Physics, IISER Thiruvananthapuram
Supervisor: Professor M. M. Shaijumon

March 2025 - April 2025

- Synthesized and characterized sodium vanadium phosphate (NVP) as a cathode for sodium-ion batteries.
- Performed structural (XRD, SEM) and electrochemical (CV, GCD, EIS) analyses to assess performance.
- Evaluated charge/discharge behavior, cycling stability, and material efficiency.
- Technical Skills:
- Battery material synthesis, electrochemical analysis, XRD/SEM, data interpretation.

Outcome: Achieved stable NVP cathode performance; presented findings supporting sodium-ion batteries for sustainable energy storage.

WINTER INTERNSHIP

Department of Physics, University of Delhi
Supervisor: Professor Govind Dayal

Dec 2024 - Jan 2025

- Advanced literature on plasmonic nanoantennas, metamaterials, and SEIRA for biomolecular analysis.
- Investigated integration of nanotechnology with bioinformatics, focusing on structural protein analysis and real-time interaction studies.
- Used computational tools for modeling and evaluation.
- Technical Skills:
- Literature review, nanotechnology concepts, computational modeling, technical communication.

Outcome: Presented report on applying nanotechnology in bioinformatics for enhanced molecular diagnostics.

Department of Biotechnology, University of Delhi
Supervisor: Professor Manish Kumar

Dec 2024 - Jan 2025

- Developed a mutation analysis tool in R and Shiny for protein sequence analysis and drug resistance prediction.
- Automated sequence translation, mutation visualization, and implemented BLAST algorithm for comparison.
- Built an interactive web application to support bioinformatic research and clinical use.
- Technical Skills:
- R programming, data visualization, sequence analysis, web app development (Shiny).

Outcome: Delivered usable bioinformatics software integrating mutation mapping, aiding research and clinical decision making

SKILLS AND EXTRACURRICULAR

- Developed strong research skills through participation in projects and laboratory work at IISER Thiruvananthapuram.
- Working knowledge of C, C++, and Python gained through coursework at IISER Thiruvananthapuram.
- Seikokai karate (Black Belt 1st dan)
- Volleyball
- Aloha (Level 3)
- Volunteered with the "Unnat Bharat Abhiyan" project at IISER TVM, tutoring underprivileged students in mathematics and science at a government school.

CONFERENCES AND SUMMER SCHOOLS

- StuFoTech Summer School, Kiel University, 2026: Presented work on SAW magnetoelastic detection of magnetic nanoparticles; received the Best Presentation Award.
- CRC 1261 Biomedical Sensing Summer School, 2026: Presented a poster on magnetic nanoparticle-based biosensing within Project B12, focusing on AC susceptibility, Delta-E, and SAW magnetoelastic sensors.

PUBLICATION

Co-author, accepted conference paper for IEEE SENSORS 2026: “Magnetic Nanoparticle Quantification Using a Delta-E Magnetoelectric Sensor”; contributed to MNP measurements and data analysis.

REFERENCES

Available on request